SRM UNIVERSITY AP

Department of Physics

Summer Internship Fellowship 2026

A report on

Quantum Error Correction

A Beginner's Introduction

Submitted by

Mr. Abhishek P G

Department of Physics

St.Aloysius College, Elthuruth

Under the guidance of

Dr. Soumyakanti Bose

Department of Physics

SRM University AP

DECLARATION

I hereby declare that the internship report entitled **"Quantum Error Correction: A Beginner's Introduction "** is an original work carried out by me during the **Summer Internship Fellowship (SIF) 2026** at the **Department of Physics, SRM University–AP**, under the guidance and supervision of **Dr. Soumyakanti Bose**. This report has been prepared based on the work completed during the internship and has not been submitted, either in whole or in part, to any other institution for the award of any degree, diploma, or other academic qualification.

Signature of Student: ______________________________

Name of Student: **ABHISHEK P G**

Affiliation: St. Aloysius College, Elthuruth

Signature of Faculty Mentor: ______________________________

Name of Mentor: **Dr. Soumyakanti Bose**

SRM University–AP
Andhra Pradesh
July 2026

Acknowledgment

I would like to express my sincere gratitude to my guide, **Dr. Soumyakanti Bose**, for his immense support, guidance, and the invaluable knowledge he shared with me throughout the course of this internship.

I am also thankful to **Dr. Ravi Kumar** for selecting me for the Summer Internship Fellowship (SIF) 2026, and to **Dr. Soumyajyoti Biswas**, Associate Professor and Head, Department of Physics, for providing the facilities and a conducive academic environment that made this work possible.

I would further like to thank **Prof. Jayaseelan**, Associate Dean, for his support, and the entire **SEAS, SRM University – AP** team for organising the Summer Internship Fellowship 2026 and making this opportunity possible.

Finally, I extend my thanks to the Department of Physics, SRM University – AP, and to St. Aloysius College, Elthuruth, for their continued support during this internship.

SRM University–AP
Andhra Pradesh
July 2026

Abstract

Quantum computation and information is a rapidly growing field, and quantum error correction (QEC) sits at its core as an essential area of research. Unlike classical bits, qubits are fragile: the no-cloning theorem forbids copying an unknown quantum state, and direct measurement collapses superposition, ruling out classical error-correction strategies such as simple repetition and majority voting. This report presents a beginner's introduction to QEC, building the subject from first principles. We begin with the classical repetition code as a point of contrast, then examine the constraints unique to quantum systems that necessitate new techniques. We develop the three-qubit bit-flip and phase-flip codes in detail, showing step by step how encoding, syndrome extraction, and correction allow errors to be diagnosed and reversed without ever measuring the encoded information directly. These two codes are then unified through concatenation in the nine-qubit Shor code, the first construction shown to protect a logical qubit against an arbitrary single-qubit error. Finally, we introduce the stabilizer formalism as a compact and systematic language for describing such codes, connecting the explicit constructions developed here to the broader framework underlying modern quantum error-correcting codes.

Contents

1 Introduction

Quantum computers are often praised for their enormous computational power. However, like every physical system, they come with their own challenges. Classical computers, which we use in our day-to-day lives, are also susceptible to errors caused by electrical noise, extreme temperatures, manufacturing imperfections, or even cosmic ray interactions. Such errors can often be detected and corrected using relatively simple techniques, such as the 3-bit repetition code.

Quantum computers, on the other hand, are significantly more vulnerable to errors because quantum information is extremely fragile. The interaction of qubits with their surrounding environment leads to the loss of quantum information, a phenomenon known as **decoherence**. Unlike classical systems, we cannot directly apply classical error-correction techniques to quantum computers. This is because quantum mechanics imposes fundamental constraints, such as the **no-cloning theorem**. and the fact that measuring a **qubit collapses its quantum state**, potentially destroying the information we wish to protect. Consequently, quantum error correction requires entirely different strategies that preserve quantum information without directly measuring or copying it.

2 Classical Error Correction

Classical computers use binary encoding $\{0, 1\}$. Whenever information travels through a noisy channel, there is a chance of a bit getting flipped. For example, if we try to send the bit '0', the channel might corrupt it into '1':

$$\underset{\text{Sender}}{\text{'0'}} \xrightarrow{\text{Noisy Channel}} \underset{\text{Receiver}}{\text{'1'}}$$

To overcome this, the simplest fix is the **repetition code** [1]: instead of sending a single bit, we encode it using 3 bits.

$$\{0, 1\} \xrightarrow{\text{3-bit encoding}} \{000, 111\} \tag{1}$$

The states 000 and 111 are called **logical codewords**. This works because, if each bit is flipped independently with some small probability p, it is far more likely that only one bit gets corrupted than that two or more bits get corrupted at once. So even if a single bit is flipped, we can still recover the original message by **majority voting**:

$$\underset{\text{Sender}}{\text{'000'}} \xrightarrow{\text{Noisy Channel}} \underset{\text{Receiver}}{\text{'010'}}$$
$$\text{majority vote} \rightarrow \text{'0'}$$

i.e., the **Hamming distance** between them is $d = 3$ so that it can detect up to $d - 1 = 2$ errors and correct up to $\frac{d-1}{2} = 1$.

3 Quantum bits - The Qubits

In place of bits in classical systems, the fundamental unit of quantum information is the **qubit**.That's the qubit state is generally can be written as follows:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \tag{2}$$

where $\alpha, \beta \in \mathbb{C}$ and it must satisfy $|\alpha|^2 + |\beta|^2 = 1$. Taking our classical error correcting method into account we can't merely copy the qubit information :

3.1 No Cloning Theorem

In classical computing copying a bit is trivial: if a state 0 is there we can simply make copies of that as 000 but quantum computing won't provide us such luxury because of *No cloning theorem* [2], which states that there is no quantum operation that can take an arbitrary, unknown quantum state and produce an independent copy of it. That is, for an unknown state $|\psi\rangle$, there is no unitary operator U that acts as

$$U |\psi\rangle = |\psi\rangle \otimes |\psi\rangle \tag{3}$$

for *every* possible $|\psi\rangle$. It is worth noting that this restriction only applies to *unknown* or arbitrary states. If we already know a qubit is in a fixed basis state, say $|0\rangle$, then preparing more qubits in that same state is trivial:

$$|0\rangle \otimes |0\rangle \otimes |0\rangle = |000\rangle \tag{4}$$

This is not really "cloning" in the quantum sense, it is just repeated preparation of a known state, so no contradiction arises here.

The trouble begins when we try to do the same thing for a superposition state $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$. If cloning were possible, we would expect

$$|\psi\rangle \otimes |\psi\rangle \otimes |\psi\rangle \tag{5}$$

which, when expanded, contains cross terms involving every combination of 0's and 1's across the three qubits, which isn't possible to detect the errors occurred.

3.2 Qubit Collapse

Whenever we try to detect errors by directly measuring the data qubits, the quantum state collapses into one of the computational basis states ($|0\rangle$ or $|1\rangle$). This destroys the superposition and the quantum information, making further quantum computation impossible.

These fundamental challenges motivated the development of quantum error-correcting codes. By encoding a logical qubit into multiple physical qubits and using syndrome measurements, these codes can efficiently detect and correct errors without collapsing the quantum state, making reliable quantum computation possible.

4 Quantum Error Correction

Quantum systems are very much prone to noise. Unlike classical systems, even minute environmental disturbances can introduce errors into quantum states, leading to the loss of quantum information. Furthermore, the no-cloning theorem prevents the direct copying of an unknown quantum state, and direct measurement causes the quantum state to collapse.
To overcome these challenges, **Quantum Error Correction (QEC)** was developed. **QEC** protects quantum information by encoding a single logical qubit into multiple physical qubits. Rather than measuring the data qubits directly, it uses syndrome measurements on ancillary qubits to identify the presence and location of errors without disturbing the encoded quantum state. Once the error is identified, the appropriate correction operation is applied to recover the original quantum information.
The simplest error correcting codes are designed to overcome to fundamental errors:

- **Bit-flip (X) errors** : corrected using the Three-Qubit Bit-Flip Code.
- **Phase-flip (Z) errors** :corrected using the Three-Qubit Phase-Flip Code.

4.1 3 Qubit Bit-Flip Code

In decoherence, there is a chance of X transformation to be done on the state which cause $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ to be:

$$X|\psi\rangle = \alpha|1\rangle + \beta|0\rangle \tag{6}$$

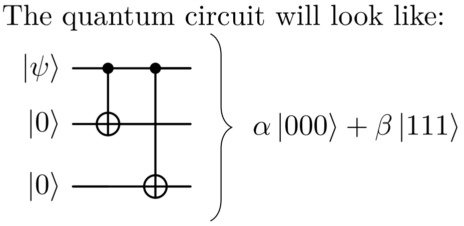
In such cases the information has different meaning to overcome this issue we need to apply this technique. For the simple understanding let's do step by step
Step 1: State Preparation
First, we prepare an entangled encoded state by applying CNOT gates to the original qubit, thereby distributing the quantum information across multiple physical qubits. This redundancy enables the detection and correction of errors without directly measuring the encoded quantum state. Suppose the original state;

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

then after applying the CNOT to two corresponding $|0\rangle$:

$$|\psi_L\rangle = \alpha|000\rangle + \beta|111\rangle$$

The quantum circuit will look like:



Step 2: Error Introduction

The second is the main part – the reason and purpose for such techniques – that is, the introduction of a bit-flip error. Suppose an X error is introduced on one of the three qubits:

$$
\begin{aligned}
X_1 II &\longrightarrow \alpha\,|100\rangle + \beta\,|011\rangle \\
IX_2 I &\longrightarrow \alpha\,|010\rangle + \beta\,|101\rangle \\
IIX_3 &\longrightarrow \alpha\,|001\rangle + \beta\,|110\rangle
\end{aligned}
$$

Step 3: Syndrome Detection

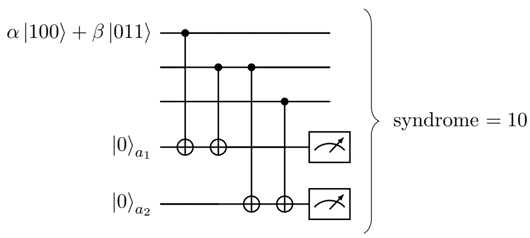
To detect which qubit was flipped without collapsing the encoded information, we introduce two ancilla qubits and use them to measure the parity between qubit pairs (1,2) and (2,3)[1]. This reveals the error location while leaving α and β undisturbed. Suppose an error occurred on qubit 1, giving the state:

$$
\begin{aligned}
\text{Signal} &= \alpha\,|0\rangle + \beta\,|1\rangle \\
\text{Code} &= \alpha\,|000\rangle + \beta\,|111\rangle \\
\text{Code+anc.} &= \alpha\,|000\rangle\,|00\rangle + \beta\,|111\rangle\,|00\rangle \\
\text{Error at 1} &= \alpha\,|100\rangle\,|00\rangle + \beta\,|011\rangle\,|00\rangle \\
\text{CNOT}_{P_1,S_1} &= \alpha\,|100\rangle\,|10\rangle + \beta\,|011\rangle\,|00\rangle \\
\text{CNOT}_{P_2,S_1} &= \alpha\,|100\rangle\,|10\rangle + \beta\,|011\rangle\,|10\rangle \\
\text{CNOT}_{P_2,S_2} &= \alpha\,|100\rangle\,|10\rangle + \beta\,|011\rangle\,|11\rangle \\
\text{CNOT}_{P_3,S_2} &= \alpha\,|100\rangle\,|10\rangle + \beta\,|011\rangle\,|10\rangle \\
&= \big(\alpha\,|100\rangle + \beta\,|011\rangle\,\big)_P\ |10\rangle_S
\end{aligned}
$$

$$S_1 = P_1 P_2, \qquad S_2 = P_2 P_3$$

$S_1 \backslash S_2$	0	1
0	$\|000\rangle\,,\ \|111\rangle$	$\|001\rangle\,,\ \|110\rangle$
1	$\|100\rangle\,,\ \|011\rangle$	$\|010\rangle\,,\ \|101\rangle$

Circuit for Syndrome Extraction



$$\text{Syndrome} = 10 \ \longrightarrow\ \text{Error on qubit 1}$$

Step 4: Error Correction

Once the syndrome is measured, it is used classically to identify which qubit was affected, and a corrective X gate is applied to that qubit to restore the original encoded state. For a syndrome of 10, indicating an error on qubit 1, we apply X_1:

$$
\begin{aligned}
\text{Before correction} &= \big(\alpha\,|100\rangle + \beta\,|011\rangle\,\big)_P\ |10\rangle_S \\
X_1\ \big(\alpha\,|100\rangle + \beta\,|011\rangle\,\big) &= \alpha\,|000\rangle + \beta\,|111\rangle \\
\text{After correction} &= \big(\alpha\,|000\rangle + \beta\,|111\rangle\,\big)_P\ |10\rangle_S
\end{aligned}
$$

[1] The choice of parity checks is not unique – any two independent stabilizer generators (e.g. Z_1Z_2, Z_1Z_3 or Z_1Z_3, Z_2Z_3) will work equally well to detect the error. Only the resulting syndrome table changes accordingly, not the underlying logic.

The physical (data) qubits are restored to the original encoded state $|\psi_L\rangle = \alpha|000\rangle + \beta|111\rangle$, while the ancillas retain the syndrome $|10\rangle_S$ – these are reset (or discarded and replaced with fresh $|00\rangle$ ancillas) before the next round of error correction.

4.2 3 Qubit Phase-Flip Code

Similar to the bit-flip (X) error, the phase-flip (Z) error is a fundamental error encountered in quantum systems due to decoherence and environmental interactions. Where when $Z|0\rangle = |0\rangle \& Z|1\rangle = -|1\rangle$ then suppose for $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ to be:

$$Z|\psi\rangle = \alpha|0\rangle - \beta|1\rangle$$

Such a type of error has no classical analog, making it harder to correct directly. The key insight is that a phase-flip error acts like a bit-flip error in the X (Hadamard) basis. Recall the Hadamard basis states [2]:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

so that $Z|+\rangle = |-\rangle$ and $Z|-\rangle = |+\rangle$ – i.e. Z flips $|+\rangle \leftrightarrow |-\rangle$ exactly as X flips $|0\rangle \leftrightarrow |1\rangle$. This means the phase-flip code can be built by first rotating into the Hadamard basis (via H gates), applying the *same* bit-flip encoding as before, and rotating back at the end.

Step 1: State Preparation
We first apply a Hadamard gate to the original qubit, then encode using CNOTs exactly as in the bit-flip code:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \xrightarrow{H} \alpha|+\rangle + \beta|-\rangle$$
$$\alpha|+\rangle + \beta|-\rangle \xrightarrow{\text{CNOT}} \alpha|{+}{+}{+}\rangle + \beta|{-}{-}{-}\rangle$$

Step 2: Error Introduction
Suppose a Z error is introduced on one of the three qubits during transmission:

$$\begin{aligned} Z_1 II &\longrightarrow \alpha|{-}{+}{+}\rangle + \beta|{+}{-}{-}\rangle \\ IZ_2 I &\longrightarrow \alpha|{+}{-}{+}\rangle + \beta|{-}{+}{-}\rangle \\ IIZ_3 &\longrightarrow \alpha|{+}{+}{-}\rangle + \beta|{-}{-}{+}\rangle \end{aligned}$$

Step 3: Syndrome Detection
To detect which qubit was phase-flipped, we again use two ancilla qubits – but since the error now lives in the X basis, we first rotate each data qubit back with a Hadamard before applying the same CNOT-based parity checks used in the bit-flip code. Suppose the error occurred on qubit 1, giving $\alpha|{-}{+}{+}\rangle + \beta|{+}{-}{-}\rangle$; after applying $H^{\otimes 3}$ this becomes $\alpha|100\rangle + \beta|011\rangle$, and the syndrome extraction proceeds exactly as before[2]:

$$S_1 = P_1 P_2, \qquad S_2 = P_2 P_3$$

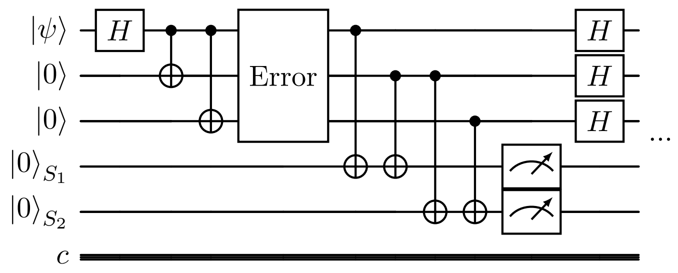
[2] As with the bit-flip code, the choice of parity checks ($S_1 = P_1P_2$, $S_2 = P_2P_3$) is not unique – only the resulting syndrome table changes.

$$\text{Syndrome} = 10 \longrightarrow \text{Phase error on qubit 1}$$

Step 4: Error Correction
Once the syndrome identifies the affected qubit, a corrective Z gate is applied to that qubit (in the original basis) to restore the encoded state:

$$Z_1 \left(\alpha \left|-++\right\rangle + \beta \left|+--\right\rangle\right) = \alpha \left|+++\right\rangle + \beta \left|---\right\rangle$$

restoring $|\psi_L\rangle = \alpha\,|+++\rangle + \beta\,|---\rangle$. A final Hadamard on each qubit ($H^{\otimes 3}$) then returns the state to the computational basis if needed, completing the correction cycle.
After completing the encoding, error detection, syndrome measurement, and recovery procedures, the complete phase-flip error correction circuit is obtained as shown below.



5 Stabilizer Codes

The bit-flip and phase-flip codes demonstrate the basic principles of quantum error correction by protecting quantum information against specific types of errors. However, as quantum systems become larger and more complex, explicitly writing all encoded logical states becomes difficult. A more systematic method is therefore required to describe quantum error-correcting codes.

The **stabilizer formalism**, introduced by Daniel Gottesman, provides such a framework. Instead of describing a quantum code by explicitly writing its encoded states, the stabilizer formalism defines the code using a set of operators called *stabilizers* [1]. These operators characterize the valid code space and enable efficient error detection without directly measuring the encoded quantum information.

5.1 Stabilizer Operator

A stabilizer is an operator S that leaves an encoded quantum state unchanged. If $|\psi\rangle$ is a valid codeword, then

$$S|\psi\rangle = |\psi\rangle. \tag{7}$$

This means that $|\psi\rangle$ is an eigenstate of the operator S with eigenvalue $+1$. Any state satisfying this condition belongs to the code space, while states that do not satisfy it are considered to contain errors.

5.2 Pauli Operators

Stabilizer codes are constructed using tensor products of the Pauli operators

$$I,\ X,\ Y,\ Z, \tag{8}$$

where

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For multi-qubit systems, tensor products of these operators are used. For example,

$$XIZ = X \otimes I \otimes Z,$$

indicates that X acts on the first qubit, the identity operator acts on the second qubit, and Z acts on the third qubit.

5.3 Example: Three-Qubit Bit-Flip Code

The three-qubit bit-flip code can be described compactly using stabilizers. The logical basis states are

$$|0_L\rangle = |000\rangle, \qquad |1_L\rangle = |111\rangle. \tag{9}$$

The stabilizer generators are

$$S_1 = Z_1 Z_2, \qquad S_2 = Z_2 Z_3. \tag{10}$$

Both logical states satisfy

$$S_1|\psi_L\rangle = |\psi_L\rangle, \qquad S_2|\psi_L\rangle = |\psi_L\rangle, \tag{11}$$

which means they belong to the code space.

5.4 Syndrome Measurement

When an error occurs, the outcomes of stabilizer measurements may change from $+1$ to -1. The collection of these measurement outcomes is called the *error syndrome*. The syndrome identifies the location of the error without revealing the encoded quantum information.

For the three-qubit bit-flip code, the syndromes are shown in Table 1.

Table 1: Error syndromes for the three-qubit bit-flip code.

Error	$S_1 = Z_1 Z_2$	$S_2 = Z_2 Z_3$	**Syndrome**
No Error	$+1$	$+1$	$(+,+)$
X_1	-1	$+1$	$(-,+)$
X_2	-1	-1	$(-,-)$
X_3	$+1$	-1	$(+,-)$

Each syndrome corresponds to a unique error. Once the syndrome is identified, the appropriate correction operator is applied to recover the original logical state. Since only the syndrome is measured, the encoded quantum information remains undisturbed.

The stabilizer formalism provides a compact and efficient description of quantum error-correcting codes and forms the basis of many important codes, including the Shor code, Steane code, and modern surface codes.

6 The Nine-Qubit Shor Code

The three-qubit codes discussed so far protect against *either* a bit-flip *or* a phase-flip error, but not both simultaneously. In 1995, Peter Shor showed that by **concatenating** the two codes – encoding first against phase-flips, then encoding each of those qubits again against bit-flips – a single logical qubit can be protected against an arbitrary single-qubit error using nine physical qubits [1].

6.1 Concept: Concatenation

We first apply the phase-flip encoding (Section 4.2) to spread $|\psi\rangle$ across three *blocks*:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle \;\longrightarrow\; \alpha\,|{+}{+}{+}\rangle + \beta\,|{-}{-}{-}\rangle$$

Each of these three qubits is then individually encoded using the bit-flip code (Section 4.1), replacing $|+\rangle$ and $|-\rangle$ with their own 3-qubit encoded blocks:

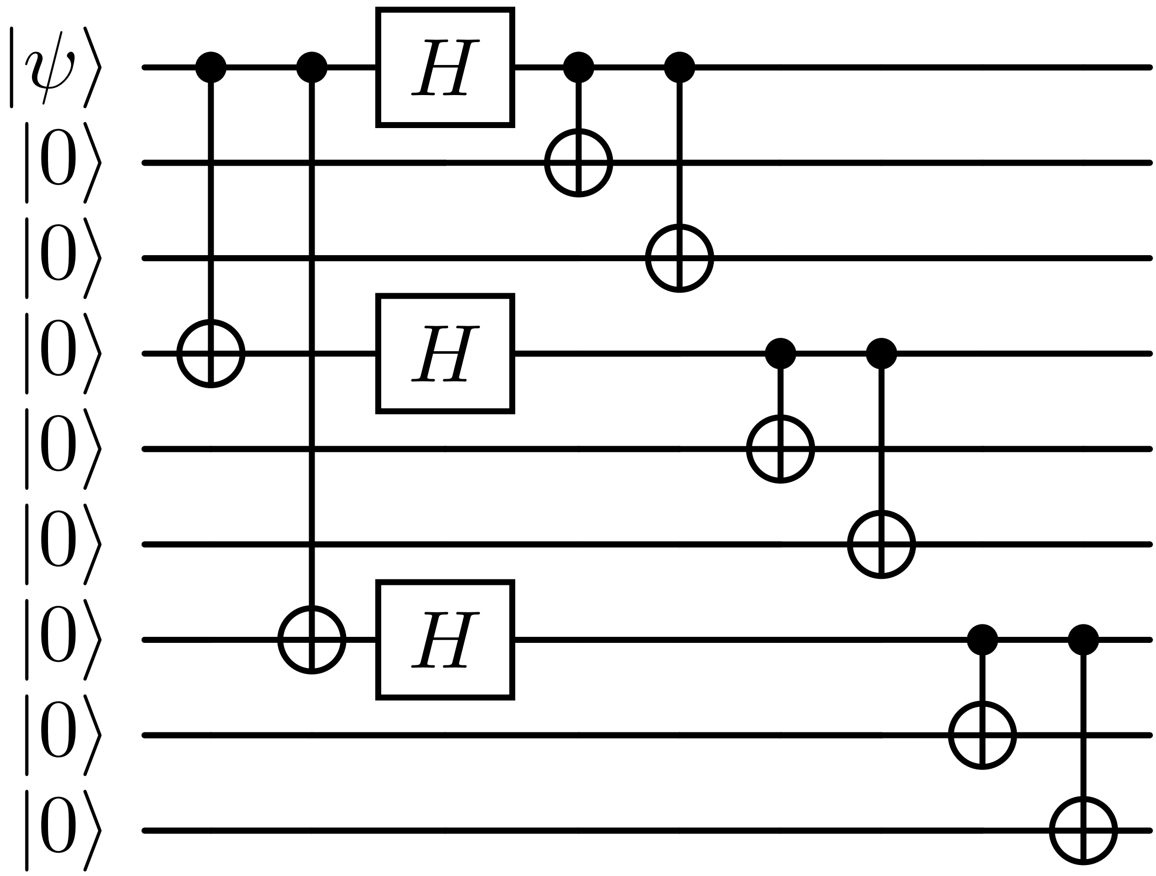
$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \;\longrightarrow\; \frac{|000\rangle + |111\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \;\longrightarrow\; \frac{|000\rangle - |111\rangle}{\sqrt{2}}$$

giving the final nine-qubit logical states:

$$|0_L\rangle = \frac{1}{2\sqrt{2}}\big(\,|000\rangle + |111\rangle\,\big)^{\otimes 3}, \qquad |1_L\rangle = \frac{1}{2\sqrt{2}}\big(\,|000\rangle - |111\rangle\,\big)^{\otimes 3}$$

6.2 Encoding Circuit

The encoding proceeds in three stages: (1) spread $|\psi\rangle$ to qubits 1, 4, and 7 via CNOTs, (2) apply H to each of these three qubits, (3) bit-flip-encode each one into its own block of three:



6.3 Stabilizers of the Shor Code

The nine-qubit code is stabilized by eight independent generators: six *within-block* Z-type stabilizers that detect bit-flips inside each triplet, and two *cross-block* X-type stabilizers that detect phase-flips between blocks.

Type	**Stabilizer**	**Detects**
Within block 1	$Z_1Z_2,\ Z_2Z_3$	bit-flip in qubits 1–3
Within block 2	$Z_4Z_5,\ Z_5Z_6$	bit-flip in qubits 4–6
Within block 3	$Z_7Z_8,\ Z_8Z_9$	bit-flip in qubits 7–9
Cross block	$X_1X_2X_3X_4X_5X_6$	phase-flip between blocks 1,2
Cross block	$X_4X_5X_6X_7X_8X_9$	phase-flip between blocks 2,3

With $9 - 1 = 8$ stabilizer generators for $n = 9$ physical qubits encoding $k = 1$ logical qubit, the syndrome from these eight measurements uniquely identifies – and thereby corrects – any single bit-flip, phase-flip, or combined (Y) error on any one of the nine physical qubits, without ever measuring or disturbing α and β directly.

6.4 Stabilizer Table (Explicit Form)

Writing out the identity operators explicitly, the eight stabilizer generators of the nine-qubit Shor code act on the physical qubits as follows:

	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8	Q_9
S_1	Z	Z	I	I	I	I	I	I	I
S_2	I	Z	Z	I	I	I	I	I	I
S_3	I	I	I	Z	Z	I	I	I	I
S_4	I	I	I	I	Z	Z	I	I	I
S_5	I	I	I	I	I	I	Z	Z	I
S_6	I	I	I	I	I	I	I	Z	Z
S_7	X	X	X	X	X	X	I	I	I
S_8	I	I	I	X	X	X	X	X	X

The first six generators (S_1–S_6) are Z-type, each acting on a pair of qubits within a single block, and detect **bit-flip** errors – exactly as in the three-qubit bit-flip code applied independently to each block. The last two generators (S_7, S_8) are X-type, each spanning two full blocks (six qubits), and detect **phase-flip** errors across blocks – analogous to the parity checks in the three-qubit phase-flip code, but acting collectively on each block as a unit rather than on individual qubits.

7 Conclusion

The immense computational power promised by quantum computers can only be realized once a fundamental obstacle is overcome: their extreme sensitivity to noise and decoherence. Protecting quantum information therefore requires a fundamentally different approach from classical error correction – one that respects the no-cloning theorem and avoids collapsing the state through direct measurement.

In this report, we traced this idea from its simplest form to a complete solution. The three-qubit bit-flip and phase-flip codes each protect against a single type of error by encoding one logical qubit into three physical qubits and using ancilla-based syndrome measurements to detect – and correct – errors without disturbing the encoded information. The nine-qubit Shor code then unifies both approaches through concatenation, protecting against an arbitrary single-qubit error and marking the first complete demonstration that quantum information can be reliably preserved.

These constructions require a careful interplay of creativity and rigorous logic, and they remain the conceptual foundation on which modern, more efficient codes – such as stabilizer codes and surface codes – are built. As the field continues to advance toward practical, fault-tolerant quantum computers, these basic principles remain the motivation driving that progress.

References

[1] Joschka Roffe, *Quantum Error Correction: An Introductory Guide*, Contemporary Physics, 60(3), 226–245, 2019.

[2] Parag K. Lala, *Quantum Computing: A Beginner's Introduction*, McGraw-Hill, India, 2020.